



Ampicillin and sulbactam: Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Contributor Disclosures

For additional information see "[Ampicillin and sulbactam: Patient drug information](#)" and "[Ampicillin and sulbactam: Pediatric drug information](#)"

For abbreviations, symbols, and age group definitions [show table](#)

Brand Names: US

Unasyn

Pharmacologic Category

Antibiotic, Penicillin

Dosing: Adult

Dosage guidance:

Dosing: Adult dosage recommendations are expressed as total grams of ampicillin/sulbactam.

Dosage forms information: Ampicillin/sulbactam is a combination product formulated in a 2:1 ratio.

Usual dosage range: IM, IV: 1.5 to 3 g every 6 hours (maximum: ampicillin/sulbactam 12 g daily) ([Ref](#)); for the treatment of infections caused by *Acinetobacter* spp., higher doses have been described ([Ref](#)).

[Collapse All](#)

Acinetobacter baumannii infection, multidrug resistant



***Acinetobacter baumannii* infection, multidrug resistant (alternative agent): IV:** 9 g every 8 hours over 4 hours **or** 27 g/day over 24 hours as a continuous infusion; alternatively, may consider 3 g every 4 hours, especially if intolerance or toxicity precludes the use of higher doses. Use as part of an appropriate combination regimen ([Ref](#)).

Anthrax, systemic, treatment

Anthrax, systemic (including meningitis), treatment (off-label use):

Note: Consult public health officials for event-specific recommendations.

IV: 3 g every 6 hours, in combination with other appropriate agents for ≥ 2 weeks; duration may be shortened and patient transitioned to oral therapy based on response and clinical judgement ([Ref](#)). Some experts suggest ≥ 3 weeks of IV combination therapy for patients with meningitis ([Ref](#)). After aerosol exposure, transition patients who are immunocompromised from treatment to postexposure prophylaxis; combined duration should total 60 days. **Note:** Administer antitoxin in addition to antibiotics for systemic anthrax ([Ref](#)).

Bite wound infection, treatment

Bite wound infection, treatment (animal or human bite) (off-label use): IV: 1.5 to 3 g every 6 hours ([Ref](#)); some experts prefer 3 g every 6 hours ([Ref](#)). Continue treatment for 1 to 2 days after resolution of infection; total duration is typically 5 to 14 days, although deep or complicated infections may require a longer duration ([Ref](#)).

Bloodstream infection

Bloodstream infection (off-label use): *For pathogen-directed therapy of susceptible organisms:*

IV: 3 g every 6 hours ([Ref](#)). Usual duration is 7 to 14 days; individualize depending on organism, source of infection, and clinical response. A 7-day duration is recommended for patients with uncomplicated Enterobacteriaceae infection who respond appropriately to antibiotic therapy ([Ref](#)).

Diabetic foot infection, moderate to severe

Diabetic foot infection, moderate to severe: IV: 3 g every 6 hours ([Ref](#)). Usual duration of therapy (including oral step-down therapy) is 2 to 4 weeks (in the absence of osteomyelitis) ([Ref](#)).

Endocarditis, treatment

Endocarditis, treatment (off-label use): *Enterococcus* (native or prosthetic valve; beta-lactamase-producing strains susceptible to aminoglycosides):

IV: 3 g every 6 hours in combination with gentamicin for 6 weeks ([Ref](#)). Some experts recommend gentamicin for the first 2 weeks of therapy ([Ref](#)).

Odontogenic soft tissue infection, pyogenic

Odontogenic soft tissue infection, pyogenic (off-label use): IV: 3 g every 6 hours; following clinical improvement, transition to oral step-down therapy and continue antibiotics until resolution, typically for a total of 7 to 14 days. Use in addition to appropriate surgical management (eg, drainage and/or extraction) ([Ref](#)).

Pelvic infections

Pelvic infections (alternative agent):

***Pelvic inflammatory disease (including tubo-ovarian abscess):* IV:** 3 g every 6 hours in combination with doxycycline. After 24 to 48 hours of sustained clinical improvement, may transition to oral therapy to complete 14 days of treatment ([Ref](#)).

***Postpartum endometritis:* IV:** 3 g every 6 hours; treat until patient is clinically improved (no fundal tenderness) and afebrile for 24 to 48 hours ([Ref](#)).

Peritonitis, treatment, peritoneal dialysis

Peritonitis, treatment, peritoneal dialysis (off-label route):

Note: Intraperitoneal administration is preferred to IV administration unless the patient has sepsis ([Ref](#)).

Continuous (with every CAPD exchange): Intraperitoneal: Loading dose: 1.5 g/L of dialysate added to first dialysate exchange; maintenance dose: 200 mg/L of dialysate with each subsequent dialysate exchange ([Ref](#)).

Duration: For patients with adequate clinical response, duration of therapy is ≥ 2 to 3 weeks depending on organism. For patients with no improvement after 5 days, remove catheter and treat with appropriate systemic antibiotics for 14 days after catheter removal ([Ref](#)).

Pneumonia



Pneumonia (off-label use):

Aspiration pneumonia, community-acquired (nonsevere): IV: 1.5 to 3 g every 6 hours, generally for 5 days (including oral step-down therapy) ([Ref](#)).

Community-acquired pneumonia: Inpatients without risk factors for *P. aeruginosa*:
IV: 3 g every 6 hours in combination with other agent(s) when appropriate. Total duration (including oral step-down therapy) is a minimum of 5 days; patients should be clinically stable with normal vital signs prior to discontinuation ([Ref](#)).

Hospital-acquired or ventilator-associated pneumonia: IV: 3 g every 6 hours, as part of a combination regimen when appropriate. Duration of therapy varies based on disease severity and response to therapy; treatment is typically given for 7 days ([Ref](#)).

Surgical prophylaxis



Surgical prophylaxis (off-label use): IV: 3 g within 60 minutes prior to surgical incision. Doses may be repeated in 2 hours if procedure is lengthy or if there is excessive blood loss. **Note:** Consider local susceptibility patterns prior to use ([Ref](#)). Postoperative prophylaxis is not recommended in clean and clean-contaminated surgeries ([Ref](#)).

Surgical site infection



Surgical site infection (eg, intestinal, GU tract, abdominal wall) (off-label use): IV: 3 g every 6 hours. Duration depends on extent and severity of infection as well as response to therapy; may switch to oral treatment when clinically improved. **Note:** Consult local susceptibility patterns prior to empiric use ([Ref](#)).

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV:

Note: Dosage recommendations are expressed as total grams of **ampicillin/sulbactam** (Ref):

Ampicillin/Sulbactam Dosage Adjustment for Kidney Impairment (Usual Dosage Range 1.5 to 3 g IV Every 6 Hours)

Traditional intermittent infusion method (administered over ≤30 minutes)		
CrCl	If the usual recommended dose is 1.5 g IV every 6 hours	If the usual recommended dose is 3 g IV every 6 hours
CrCl ≥30 mL/minute	No dosage adjustment necessary	No dosage adjustment necessary
CrCl 15 to 29 mL/minute	1.5 g IV every 12 hours	3 g IV every 12 hours
CrCl <15 mL/minute	1.5 g IV every 24 hours	3 g IV every 24 hours

Ampicillin/Sulbactam Dosage Adjustment Recommendations for Kidney Impairment when Treating Multidrug-resistant *A. baumannii* Infection^a

	Traditional intermittent infusion method (administered over ≤30 minutes)	Extended infusion method (over 4 hours)	Continuous infusion method (over 24 hours) ^a
CrCl	If the usual recommended dose is 3 g IV every 4 hours administered over ≤30 minutes	If the usual recommended dose is 9 g IV every 8 hours administered over 4 hours	If the usual recommended dose is 27 g IV administered over 24 hours
CrCl ≥90 to 130 mL/minute	No dosage adjustment necessary ^b	No dosage adjustment necessary ^{c,d}	No dosage adjustment necessary
CrCl 60 to <90 mL/minute	No dosage adjustment necessary ^b	6 g IV every 8 hours over 4 hours ^{c,d}	18 g IV over 24 hours ^c
CrCl 30 to <60 mL/minute	3 g IV every 6 hours ^b	3 g IV every 6 hours over 4 hours ^{c,d}	12 g IV over 24 hours ^c
CrCl 15 to 29 mL/minute	3 g IV every 8 hours ^b	3 g IV every 8 hours over 4 hours ^{c,d}	9 g IV over 24 hours ^c
CrCl <15 mL/minute	3 g IV every 12 hours ^b	3 g IV every 12 hours over 4 hours ^e	6 g IV over 24 hours ^e

	Traditional intermittent infusion method (administered over ≤30 minutes)	Extended infusion method (over 4 hours)	Continuous infusion method (over 24 hours)^a
--	---	--	---

^a The proposed renal dose adjustments are aimed at achieving pharmacokinetic/pharmacodynamic targets based on in-vitro or pharmacokinetic modeling. Clinical data are not available.

^b Expert opinion derived from Yokoyama 2015. **Note:** Doses presented in Yokoyama 2015 are in grams of sulbactam; the doses listed above are multiplied by 3 to account for both the ampicillin and sulbactam components.

^c Expert opinion derived from Jaruratanasirikul 2019. **Note:** Doses presented in Jaruratanasirikul 2019 are in grams of sulbactam; the doses listed above are multiplied by 3 to account for both the ampicillin and sulbactam components.

^d Lower doses may be sufficient for patients with lower albumin concentrations (eg, 1.7 to 2.4 g/dL) and isolates with lower minimum inhibitory concentrations. See Jaruratanasirikul 2019 for more patient-specific recommendations.

^e Expert opinion.

Augmented renal clearance (measured urinary CrCl ≥130 mL/minute/1.73 m²):

Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Young patients (<55 years of age) admitted post trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients ([Ref](#)).

Note: Dosage recommendations are expressed as total grams of **ampicillin/sulbactam**.

Standard dosage adjustment recommendations (usual dose 1.5 to 3 g IV every 6 hours):

IV: *Traditional intermittent infusion (administered over ≤30 minutes):* 1.5 to 3 g every 4 to 6 hours ([Ref](#)).

Dosage adjustment recommendations for multidrug-resistant *A. baumannii* infection:

IV: *Continuous infusion (over 24 hours):* 27 g IV administered over 24 hours. ([Ref](#)) **Note:** 36 g IV every 24 hours has also been modeled and may be necessary in cases with elevated minimum inhibitory concentrations; no clinical data available ([Ref](#)).

Hemodialysis, intermittent (thrice weekly): Dialyzable (39% to 63%) ([Ref](#)):

Note: Dosage recommendations are expressed as total grams of **ampicillin/sulbactam**.

Standard dosage adjustment recommendations (usual dose 1.5 to 3 g IV every 6 hours):

IV: *Traditional intermittent infusion (administered over ≤30 minutes):* 1.5 to 3 g every 12 to 24 hours; administer after dialysis when scheduled dose falls on dialysis days ([Ref](#)).

Dosage adjustment recommendations for multidrug-resistant *A. baumannii* infection:

IV: Intermittent, extended or continuous infusion: Dose as for CrCl <15 mL/minute as presented in the *A. baumannii* dose adjustment table ([Ref](#)). On dialysis days, if possible, administer one of the doses after dialysis ([Ref](#)).

Peritoneal dialysis:

Note: Dosage recommendations are expressed as total grams of **ampicillin/sulbactam**.

Standard dosage adjustment recommendations (usual dose 1.5 to 3 g IV every 6 hours):

IV: Traditional intermittent infusion (administered over ≤ 30 minutes): 1.5 g every 12 hours or 3 g every 24 hours ([Ref](#)).

Dosage adjustment recommendations for multidrug-resistant *A. baumannii* infection:

IV: Intermittent, extended or continuous infusion: Dose as for CrCl <15 mL/minute as presented in the *A. baumannii* dose adjustment table ([Ref](#)).

CRRT: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent flow rates of 20 to 25 mL/kg/hour (or ~1,500 to 3,000 mL/hour) unless otherwise noted. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection) and consideration of initial loading doses. Close monitoring of response and adverse reactions (eg, neurotoxicity) due to drug accumulation is important.

Note: Dosage recommendations are expressed as total grams of **ampicillin/sulbactam**.

Standard dosage adjustment recommendations (usual dose 1.5 to 3 g IV every 6 hours):

IV: Traditional intermittent infusion (administered over ≤ 30 minutes): 3 g every 8 to 12 hours ([Ref](#)).

Dosage adjustment recommendations for multidrug-resistant *A. baumannii* infection:

IV: Intermittent, extended or continuous infusion: Dose as for CrCl 30 to <60 mL/minute as presented in the *A. baumannii* dose adjustment table ([Ref](#)).

PIRRT (eg, sustained, low-efficiency diafiltration): Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection) and consideration of initial loading doses. Close monitoring of response and adverse reactions (eg, neurotoxicity) due to drug accumulation is important.

Note: Dosage recommendations are expressed as total grams of **ampicillin/sulbactam**.

Standard dosage adjustment recommendations (usual dose 1.5 to 3 g IV every 6 hours):

IV: *Traditional intermittent infusion (administered over ≤ 30 minutes):* Initial: 3 g followed by 1.5 to 3 g every 8 to 12 hours. Where possible, give one dose after PIRRT session ([Ref](#)).

Dosage adjustment recommendations for multidrug-resistant *A. baumannii* infection:

IV: *Intermittent, extended or continuous infusion:*

PIRRT days: Dose as for CrCl 30 to <60 mL/minute as presented in the *A. baumannii* dose adjustment table ([Ref](#)).

Non-PIRRT days: Dose as for CrCl <15 mL/minute as presented in the *A. baumannii* dose adjustment table ([Ref](#)).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST, Arun Jesudian, MD; Sasan Sakiani, MD.

Liver impairment prior to treatment initiation:

Child-Turcotte-Pugh class A through C: No dosage adjustment necessary ([Ref](#)).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "[Ampicillin and sulbactam: Pediatric drug information](#)")

Dosage guidance:

Dosing: Dosage recommendations are expressed as mg of the **ampicillin** component.

Dosage forms information : Unasyn (ampicillin/sulbactam) is a combination product formulated in a 2:1 ratio (eg, each 3 g vial contains 2 g of ampicillin and 1 g of sulbactam); review dosing units carefully.

General dosing:

Infants, Children, and Adolescents: IV, IM: 100 to 200 mg ampicillin/kg/day divided every 6 hours; usual maximum dose: 2,000 mg ampicillin/dose ([Ref](#)). Higher doses are necessary in some instances; see specific indications. **Note:** Higher dosing and alternative dosing strategies (eg, extended infusions, continuous infusions) have also been used to treat *Acinetobacter baumannii* infections in adults ([Ref](#)). [Collapse All](#)

Endocarditis, treatment

Endocarditis, treatment: Limited data available: Children and Adolescents: IV: 200 to 300 mg ampicillin/kg/day divided every 4 to 6 hours; maximum dose: 2,000 mg ampicillin/dose. Recommended combination therapy and duration vary by causative pathogen; treatment duration is at least 4 to 6 weeks ([Ref](#)).

Intra-abdominal infection, complicated, treatment

Intra-abdominal infection, complicated, treatment: Note: Not recommended for empiric treatment due to high rates of *Escherichia coli* resistance ([Ref](#)).

Infants, Children, and Adolescents: IV: 200 mg ampicillin/kg/day divided every 6 hours ([Ref](#)).

Meningitis

Meningitis: Limited data available: **Note:** Experience is limited; not included in IDSA meningitis guidelines ([Ref](#)).

Infants, Children, and Adolescents: IV: 400 mg ampicillin/kg/day in divided doses every 4 to 6 hours; maximum dose: 2,000 mg ampicillin/dose ([Ref](#)).

Osteoarticular infection

Osteoarticular infection: Limited data available: Infants, Children, and Adolescents: IV: 200 mg ampicillin/kg/day in divided doses every 6 hours; maximum dose: 2,000 mg ampicillin/dose ([Ref](#)).

Pelvic inflammatory disease

Pelvic inflammatory disease (alternative agent): Adolescents: IV: 2,000 mg ampicillin every 6 hours in combination with doxycycline. After 24 to 48 hours of sustained clinical improvement, may transition to oral therapy to complete 14 days of treatment ([Ref](#)).

Rhinosinusitis, severe infection requiring hospitalization

Rhinosinusitis, severe infection requiring hospitalization: Limited data available: Children and Adolescents: IV: 200 to 400 mg ampicillin/kg/day divided every 6

hours for 10 to 14 days; maximum dose: 2,000 mg ampicillin/dose ([Ref](#)).

Skin and soft tissue infection

Skin and soft tissue infection: Children and Adolescents: IV: 200 mg ampicillin/kg/day divided every 6 hours for up to 14 days; maximum dose: 2,000 mg ampicillin/dose.

Surgical prophylaxis

Surgical prophylaxis: Limited data available: Children and Adolescents: IV: 50 mg ampicillin/kg/**dose** within 60 minutes prior to procedure; may repeat in 2 hours if lengthy procedure or excessive blood loss; maximum dose: 2,000 mg ampicillin/dose ([Ref](#)).

Dosing: Kidney Impairment: Pediatric

Children and Adolescents: IV:

CrCl ≥ 30 mL/minute/1.73 m²: No dosage adjustment required.

CrCl 15 to 29 mL/minute/1.73 m²: Administer every 12 hours.

CrCl 5 to 14 mL/minute/1.73 m²: Administer every 24 hours.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported in adults. Also see Ampicillin.

>10%: Local: Pain at injection site (IM: 16%; IV: 3%)

1% to 10%:

Cardiovascular: Phlebitis (1%), thrombophlebitis (3%)

Dermatologic: Skin rash (<2%)

Gastrointestinal: Diarrhea (3%)

<1%:

Cardiovascular: Chest pain, edema, substernal pain

Dermatologic: Erythema of skin, pruritus

Gastrointestinal: Abdominal distention, flatulence, glossitis, mucous membrane bleeding, nausea, vomiting

Genitourinary: Dysuria, urinary retention

Hypersensitivity: Facial swelling

Infection: Candidiasis

Nervous system: Chills, fatigue, headache, malaise

Respiratory: Epistaxis, pharyngeal edema

Frequency not defined:

Endocrine & metabolic: Decreased serum albumin, decreased serum total protein, increased lactate dehydrogenase

Genitourinary: Hematuria

Hematologic & oncologic: Basophilia, decreased hematocrit, decreased hemoglobin, decreased neutrophils, decreased platelet count, decreased red blood cells, decreased white blood cell count, eosinophilia, lymphocytopenia, lymphocytosis, monocytosis, thrombocytosis

Hepatic: Increased serum alanine aminotransferase, increased serum alkaline phosphatase, increased serum aspartate aminotransferase

Renal: Casts in urine (hyaline), increased blood urea nitrogen, increased serum creatinine

Postmarketing:

Dermatologic: Acute generalized exanthematous pustulosis, bullous dermatitis (linear IgA), erythema multiforme, exfoliative dermatitis, Stevens-Johnson syndrome, toxic epidermal necrolysis, urticaria

Endocrine & metabolic: Hypokalemia

Gastrointestinal: Abdominal pain, cholestasis, *Clostridioides difficile*-associated diarrhea, dyspepsia, gastritis, melanoglossia, melena, stomatitis

Hematologic & oncologic: Agranulocytosis, hemolytic anemia, immune thrombocytopenia, positive direct Coombs test

Hepatic: Cholestatic hepatitis, cholestatic jaundice, hepatitis, hyperbilirubinemia, jaundice

Hypersensitivity: Drug reaction with eosinophilia and systemic symptoms, hypersensitivity reaction (including anaphylaxis, angioedema, severe hypersensitivity reaction)

Local: Injection-site reaction

Nervous system: Dizziness, seizure

Neuromuscular & skeletal: Arthralgia

Renal: Interstitial nephritis

Respiratory: Dyspnea

Contraindications

Hypersensitivity (eg, anaphylaxis or Stevens-Johnson syndrome) to ampicillin, sulbactam, or to other beta-lactam antibacterial drugs (eg, penicillins, cephalosporins), or any component of the formulations; history of cholestatic jaundice or hepatic dysfunction associated with ampicillin/sulbactam

Warnings/Precautions

Concerns related to adverse effects:

- Anaphylactoid/hypersensitivity reactions: Serious and occasionally severe or fatal hypersensitivity (anaphylactic) reactions have been reported in patients on penicillin therapy, especially with a history of beta-lactam hypersensitivity or a history of sensitivity to multiple allergens. Patients with a history of penicillin hypersensitivity have experienced severe reactions when treated with cephalosporins. Before initiating therapy, carefully investigate previous penicillin, cephalosporin, or other allergen hypersensitivity. If an allergic reaction occurs, discontinue and institute appropriate therapy.
- Hepatic dysfunction: Hepatitis and cholestatic jaundice have been reported (including fatalities). Toxicity is usually reversible. Monitor hepatic function at regular intervals in patients with hepatic impairment.
- Rash: Appearance of a rash should be carefully evaluated to differentiate a nonallergic ampicillin rash from a hypersensitivity reaction; rash occurs in 5% to 10% of children and is a generalized dull red, maculopapular rash, generally appearing 3-14 days after the start of therapy. It normally begins on the trunk and spreads over most of the body. It may be most intense at pressure areas, elbows, and knees.
- Superinfection: Prolonged use may result in fungal or bacterial superinfection, including *C. difficile*-associated diarrhea (CDAD) and pseudomembranous colitis; CDAD has

been observed >2 months postantibiotic treatment.

Disease-related concerns:

- Infectious mononucleosis: A high percentage of patients with infectious mononucleosis have developed rash during therapy; ampicillin-class antibacterials are not recommended in these patients.
- Renal impairment: Use with caution in patients with renal impairment; dosage adjustment recommended.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution Reconstituted, Injection [preservative free]:

Unasyn: 3 g: Ampicillin 2 g and sulbactam 1 g (1 ea); 1.5 g: Ampicillin 1 g and sulbactam 0.5 g (1 ea)

Generic: 1.5 g: Ampicillin 1 g and sulbactam 0.5 g (1 ea); 3 g: Ampicillin 2 g and sulbactam 1 g (1 ea)

Solution Reconstituted, Intravenous:

Generic: 15 g: Ampicillin 10 g and sulbactam 5 g (1 ea [DSC])

Solution Reconstituted, Intravenous [preservative free]:

Unasyn: 15 g: Ampicillin 10 g and sulbactam 5 g (1 ea)

Generic: 1.5 g: Ampicillin 1 g and sulbactam 0.5 g (1 ea [DSC]); 15 g: Ampicillin 10 g and sulbactam 5 g (1 ea); 3 g: Ampicillin 2 g and sulbactam 1 g (1 ea)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (reconstituted) (Ampicillin-Sulbactam Sodium Injection)

1.5 (1-0.5) g (per each): \$2.81 - \$7.98

3 (2-1) g (per each): \$3.79 - \$17.10

Solution (reconstituted) (Ampicillin-Sulbactam Sodium Intravenous)

3 (2-1) g (per each): \$4.37

15 (10-5) g (per each): \$27.46 - \$76.80

Solution (reconstituted) (Unasyn Injection)

1.5 (1-0.5) g (per each): \$9.25

3 (2-1) g (per each): \$17.47

Solution (reconstituted) (Unasyn Intravenous)

15 (10-5) g (per each): \$87.37

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Administration: Adult

Administer around-the-clock to promote less variation in peak and trough serum levels.

IV: Administer by slow injection over 10 to 15 minutes or as an IV infusion over 15 to 30 minutes. For some indications (eg, *Acinetobacter* infections), total daily dose may be administered over 24 hours as a continuous infusion ([Ref](#)). Ampicillin and gentamicin should not be mixed in the same IV tubing.

Some penicillins (eg, ampicillin, carbenicillin, ticarcillin, and piperacillin) have been shown to inactivate aminoglycosides *in vitro*. This has been observed to a greater extent with tobramycin and gentamicin, while amikacin has shown greater stability against inactivation. Concurrent Y-site administration should be avoided.

IM: Inject deep IM into large muscle mass; a concentration of 375 mg/mL ampicillin/sulbactam (250 mg ampicillin/125 mg sulbactam per mL) is recommended; may be diluted in sterile water or lidocaine 0.5% or lidocaine 2% for IM administration.

Intraperitoneal (off-label route): May administer continuously (with every exchange) ([Ref](#)).

Preparation for Administration: Adult

IV:

Single dose vials: Reconstitute with sterile water for injection (SWFI); use 3.2 mL for the 1.5 g vial size and 6.4 mL for the 3 g vial size, yielding a 375 (250-125) mg/mL concentration. Sodium chloride 0.9% (NS) is the diluent of choice for IV infusion use.

Single-dose piggyback bottles: Reconstitute piggyback bottles with a compatible infusion solution, yielding a concentration between 3 (2-1) mg/mL to 45 (30-15) mg/mL depending on the infusion solution used (see Stability).

Pharmacy Bulk package: Reconstitute the Pharmacy Bulk Package with 92 mL of SWFI or NS. The diluent should be added in two portions. The first 50 mL should be added and the vial should be shaken to dissolve the powder. Then add the remaining 42 mL of the solution and shake the vial again, yielding a concentration of approximately 150 (100-50) mg/mL. The manufacturer states that after reconstitution, the stopper of the concentrated solution vial should be penetrated only once using a sterile transfer device or dispensing set. The Pharmacy Bulk Package should NOT be used for direct administration.

ADD-Vantage vials: See product labeling for reconstitution and preparation details.

IM: Reconstitute with SWFI **or** 0.5% or 2% lidocaine hydrochloride injection; use 3.2 mL for the 1.5 g vial size and 6.4 mL for the 3 g vial size, yielding a 375 (250-125) mg/mL concentration.

Administration: Pediatric

Parenteral:

IM: Administer by deep IM injection.

IV: Administer by slow IV injection over 10 to 15 minutes or by intermittent IV infusion over 15 to 30 minutes. Avoid infusing concomitantly with aminoglycosides if feasible; consult drug interactions database for more information.

Preparation for Administration: Pediatric

Parenteral:

IV:

Single-dose piggyback bottles: Reconstitute with compatible diluent (eg, SWFI, NS) directly in bottle to desired concentration of 3 to 45 mg/mL of total ampicillin/sulbactam (2 to 30 mg/mL of ampicillin and 1 to 15 mg/mL of sulbactam).

Single-dose vials: Reconstitute 1.5 g vial with 3.2 mL SWFI or 3 g vial with 6.4 mL SWFI; resultant concentration: 375 mg/mL of total ampicillin/sulbactam (ie, 250 mg/mL of

ampicillin and 125 mg/mL of sulbactam). Further dilute with a compatible solution to a final concentration of 3 to 45 mg/mL of total ampicillin/sulbactam (2 to 30 mg/mL of ampicillin and 1 to 15 mg/mL of sulbactam).

Pharmacy bulk vials: Reconstitute 15 g vial with 92 mL of compatible solution (eg, SWFI, NS); resultant concentration: ~150 mg/mL of total ampicillin/sulbactam (ie, 100 mg/mL of ampicillin and 50 mg/mL of sulbactam). Further dilute with a compatible solution to a final concentration of 3 to 45 mg/mL of total ampicillin/sulbactam (2 to 30 mg/mL of ampicillin and 1 to 15 mg/mL of sulbactam).

ADD-Vantage vials: Assemble vial for dilution with system-appropriate NS container only.

IM: Reconstitute with SWFI or lidocaine (0.5% or 2%); reconstitute 1.5 g vial with 3.2 mL of diluent or 3 g vial with 6.4 mL of diluent; resultant concentration: 375 mg/mL of total ampicillin/sulbactam (ie, 250 mg/mL of ampicillin and 125 mg/mL of sulbactam).

Administer within 1 hour of preparation.

Storage/Stability

Prior to reconstitution, store at 20°C to 25°C (68°F to 77°F).

IM: Concentration of 375 mg/mL (250 mg ampicillin/125 mg sulbactam) should be used within 1 hour after reconstitution.

Intermittent IV infusion: Refer to manufacturer's labeling for specific storage instructions after reconstitution and dilution (varies by concentration and diluent).

Use: Labeled Indications

Bacterial infections: Treatment of skin and skin structure, intra-abdominal, and gynecological infections caused by susceptible bacteria; spectrum is that of ampicillin plus organisms producing beta-lactamases such as *Staphylococcus aureus*, *Haemophilus influenzae*, *Escherichia coli*, *Klebsiella*, *Acinetobacter*, *Enterobacter*, and anaerobes.

Use: Off-Label: Adult

Anthrax, systemic; Bite wound infection, treatment (animal or human bite); Bloodstream infection; Endocarditis, treatment; Odontogenic soft tissue infection, pyogenic; Pneumonia; Surgical prophylaxis; Surgical site infections (eg, intestinal, GU tract, abdominal wall)

Metabolism/Transport Effects

Refer to individual components.

Drug Interactions

(For additional information: [Launch drug interactions program](#))

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acemetacin: May increase serum concentrations of Penicillins. *Risk C: Monitor*

Aescin: Antibiotics may increase serum concentrations of Aescin. Management: Consider alternatives to this combination as aescin product labeling states coadministration with antibiotics is not recommended. *Risk D: Consider Therapy Modification*

Allopurinol: May increase hypersensitivity effects of Ampicillin. *Risk C: Monitor*

Aminoglycosides: Penicillins may decrease serum concentrations of Aminoglycosides. Primarily associated with extended spectrum penicillins, and patients with renal dysfunction. *Risk C: Monitor*

Atenolol: Ampicillin may decrease bioavailability of Atenolol. *Risk C: Monitor*

Bacillus Calmette Guerin (BCG) (Intravesical): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Intravesical). *Risk X: Avoid*

Bacillus Calmette Guerin (BCG) (Percutaneous): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Percutaneous). *Risk C: Monitor*

Bacillus clausii: Antibiotics may decrease therapeutic effects of Bacillus clausii. Management: Bacillus clausii should be taken in between antibiotic doses during concomitant therapy. *Risk D: Consider Therapy Modification*

Chloroquine: May decrease serum concentrations of Ampicillin. Management: Separate the administration of ampicillin and chloroquine by at least 2 hours to minimize any potential negative impact of chloroquine on ampicillin bioavailability. *Risk D: Consider Therapy Modification*

Cholera Vaccine: Antibiotics may decrease therapeutic effects of Cholera Vaccine. Management: Avoid cholera vaccine in patients receiving systemic antibiotics, and within 14 days following the use of oral or parenteral antibiotics. *Risk X: Avoid*

Dichlorphenamide: Penicillins may increase hypokalemic effects of Dichlorphenamide. *Risk C: Monitor*

Fecal Microbiota (Live) (Oral): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Oral). *Risk X: Avoid*

Fecal Microbiota (Live) (Rectal): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Rectal). *Risk X: Avoid*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): Antibiotics may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Khat: May decrease serum concentrations of Ampicillin. Management: Consider administering ampicillin 2 hours after khat chewing to avoid reductions in ampicillin bioavailability. *Risk D: Consider Therapy Modification*

Lactobacillus and Estriol: Antibiotics may decrease therapeutic effects of Lactobacillus and Estriol. *Risk C: Monitor*

Lanthanum: May decrease serum concentrations of Ampicillin. Management: Administer oral ampicillin at least two hours before or after lanthanum. *Risk D: Consider Therapy Modification*

Lindane: Penicillins may increase neuroexcitatory and/or seizure-potentiating effects of Lindane. *Risk C: Monitor*

Methotrexate: Penicillins may increase serum concentrations of Methotrexate. *Risk C: Monitor*

Mycophenolate: Antibiotics may decrease active metabolite exposure of Mycophenolate. Specifically, concentrations of mycophenolic acid (MPA) may be reduced. *Risk C: Monitor*

Probenecid: May increase serum concentrations of Sulbactam. Management: Coadministration of probenecid with sulbactam/durlobactam and sulbactam/amoxicillin (a product available outside of the US) is not recommended, but no specific actions are recommended for other sulbactam-containing products. *Risk D: Consider Therapy Modification*

Sodium Benzoate: Penicillins may decrease therapeutic effects of Sodium Benzoate. *Risk C: Monitor*

Sodium Picosulfate: Antibiotics may decrease therapeutic effects of Sodium Picosulfate. Management: Consider using an alternative product for bowel cleansing prior to a colonoscopy in patients who have recently used or are concurrently using an antibiotic. *Risk D: Consider Therapy Modification*

Tegoprazan: May decrease serum concentrations of Ampicillin. *Risk C: Monitor*

Tetracyclines: May decrease therapeutic effects of Penicillins. *Risk C: Monitor*

Typhoid Vaccine: Antibiotics may decrease therapeutic effects of Typhoid Vaccine. Only the live attenuated Ty21a strain is affected. Management: Avoid use of live attenuated

typhoid vaccine (Ty21a) in patients being treated with systemic antibacterial agents. Postpone vaccination until 3 days after cessation of antibiotics and avoid starting antibiotics within 3 days of last vaccine dose. *Risk D: Consider Therapy Modification*

Vitamin K Antagonists: Penicillins may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Pregnancy Considerations

Both ampicillin and sulbactam cross the placenta (Foulds 1986; Maberry 1992).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of ampicillin/sulbactam may be altered (Chamberlain 1993; Foulds 1986).

As a class, penicillin antibiotics are widely used in pregnant patients. Based on available data, penicillin antibiotics are generally considered compatible for use during pregnancy (Ailes 2016; Bookstaver 2015; Crider 2009; Damkier 2019; Lamont 2014; Muanda 2017a; Muanda 2017b).

Recommendations for using ampicillin/sulbactam in the management of *Bacillus anthracis* during pregnancy are available. Maternal infection with *B. anthracis* may cause preterm labor, fetal infection, fetal distress, or fetal loss. Maternal death may also occur.

Ampicillin/sulbactam is a first line option for the treatment of systemic anthrax (with or without CNS involvement) during pregnancy. The dose of ampicillin/sulbactam in pregnant and postpartum patients is the same as in nonpregnant adults (CDC [Bower 2023]; Meaney-Delman 2014).

Untreated intra-amniotic infection (chorioamnionitis) may lead to adverse pregnancy outcomes (including pneumonia, meningitis, and sepsis) in the newborn. Maternal complications may include postpartum uterine atony with hemorrhage, endometritis, peritonitis, sepsis, or adult respiratory distress syndrome. Ampicillin/sulbactam is an alternative option for the treatment of intra-amniotic infection (ACOG 2017).

Antibiotic prophylaxis is recommended prior to all cesarean deliveries unless the patient is already receiving an appropriate antibiotic. A single dose of a targeted antibiotic administered within 60 minutes prior to the delivery is recommended; ampicillin/sulbactam has been evaluated for this purpose, although other antibiotics may be preferred (consult current recommendations) (ACOG 2018).

Breastfeeding Considerations

Ampicillin and sulbactam are present in breast milk.

A review article notes the exposure of ampicillin and sulbactam to a breastfeeding infant would be ~1% to 2% of a typical adult dose (Foulds 1986).

Recommendations for using ampicillin/sulbactam in the management of *B. anthracis* in breastfeeding patients are the same as in pregnancy. Exposure to anthrax is not considered a contraindication to breastfeeding. However, if there are active cutaneous lesions on the breast, contact with the infant should be avoided and feeding from the affected breast should not occur until >48 hours of appropriate antibiotic therapy (CDC [Bower 2023]; Meaney-Delman 2014).

The manufacturer recommends that caution be used if administering to breastfeeding patients. Ampicillin is considered compatible with breastfeeding when used in usual recommended doses. In general, antibiotics that are present in breast milk may cause nondose-related modification of bowel flora. Monitor infants for GI disturbances (WHO 2002).

Also refer to the Ampicillin monograph.

Dietary Considerations

Some products may contain sodium.

Monitoring Parameters

With prolonged therapy, monitor hematologic, renal, and hepatic function; monitor for signs of anaphylaxis during first dose. In patients with preexisting hepatic impairment, monitor hepatic function at regular intervals.

Mechanism of Action

Inhibits bacterial cell wall synthesis by binding to one or more of the penicillin-binding proteins (PBPs) which in turn inhibits the final transpeptidation step of peptidoglycan synthesis in bacterial cell walls, thus inhibiting cell wall biosynthesis. Bacteria eventually lyse due to ongoing activity of cell wall autolytic enzymes (autolysins and murein hydrolases) while cell wall assembly is arrested. The addition of sulbactam, a beta-lactamase inhibitor, to ampicillin extends the spectrum of ampicillin to include some beta-lactamase-producing organisms.

Pharmacokinetics (Adult Data Unless Noted)

Ampicillin: See Ampicillin monograph.

Sulbactam:

Distribution: Widely distributed to bile, blister, and tissue fluids; poor penetration into CSF with uninflamed meninges; higher concentrations attained with inflamed meninges.

V_d :

Children ≤ 12 years: 0.34 ± 0.12 L/kg (Nahata 1999).

Adults: 0.36 L/kg (Foulds 1986).

Protein binding: 38%.

Half-life elimination: Children ≤ 12 years (normal renal function): Mean: 0.81 ± 0.12 hours (Nahata 1999); Adults (normal renal function): 1 to 1.3 hours; **Note:** Elimination kinetics of both ampicillin and sulbactam are similarly affected in patients with renal impairment, therefore, the blood concentration ratio is expected to remain constant regardless of renal function.

Excretion: Urine (~75% to 85% as unchanged drug) within 8 hours.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Anti-infective considerations:

Parameters associated with efficacy:

Ampicillin: See Ampicillin monograph.

Sulbactam (in combination with ampicillin):

Time dependent; associated with time free drug concentration (fT) > minimum inhibitory concentration (MIC):

Acinetobacter baumannii: Goal: $\geq 40\%$ to 60% $fT > MIC$ (bactericidal) (Yokoyama 2014; Yokoyama 2015).

Expected drug exposure in patients with normal renal function:

Children ≤ 12 years of age: C_{max} (peak): IV:

15- to 40-minute infusion, steady state: Ampicillin 26.7 to 53.3 mg/kg/dose and sulbactam 13.3 to 26.7 mg/kg/dose every 6 hours: Ampicillin: 177 to 200 mg/L; sulbactam: 81.9 to 102 mg/L (Nahata 1999).

Adults: C_{max} (peak):

Note: Adult doses are expressed as the combined amount of ampicillin and sulbactam.

IV: 15-minute infusion, single dose:

1.5 g: Ampicillin: 40 to 71 mg/L; sulbactam: 21 to 40 mg/L.

3 g: Ampicillin: 109 to 150 mg/L; sulbactam: 48 to 88 mg/L.

IM:

1.5 g: Ampicillin: 8 to 37 mg/L; sulbactam: 6 to 24 mg/L.

Patient Counseling Points

(For additional information see ["Ampicillin and sulbactam: Patient drug information"](#))

What is this drug used for?

- It is used to treat bacterial infections.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

For all uses of this drug:

- Diarrhea

Injection (if given in the muscle):

- Injection site pain

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- C diff–associated diarrhea (CDAD) like if you have stomach pain, cramps, or very loose, watery, or bloody stools
- Liver problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Severe skin reactions; red, swollen, blistered, or peeling skin; red or irritated eyes; sores in the mouth, throat, nose, eyes, genitals, or any areas of skin; fever; chills; body aches; shortness of breath; or swollen glands
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country



For country code abbreviations ([show table](#))

(AE) United Arab Emirates: Unasyn; (AR) Argentina: Aminoxidin sulbactam | Ampicilina sulbactam drawer | Ampicilina sulbactam morgan | Ampicilina sulbactam norgreen | Ampicilina sulbactam pharmavial | Ampicilina sulbactam veinfar | Ampigen sb | Decilina compuesta | Decilina compuesto | Prixin | Trifacilina; (AT) Austria: Ampicillin/Sulbactam AptaPharma | Ampicillin/sulbactam astro | Unasyn; (BG) Bulgaria: Ampiplus | Ampisulcillin | Unasyn; (BR) Brazil: Ampctan | Ampicilina sodica + sulbactam sodica | Ampicilina sodica + sulbactam sodico | Libractan | Sulbacter | Ulctam | Unasyn; (CL) Chile: Ampicilina+sulbactam | Unasyna; (CN) China: Ampicillin sodium and sulbactam sodium | Ampicillin/sulbact | An te xian | Bei shu lin | Betamp | Er ye qiang | Er ye shu | Hai fu bo | Jia sai | Kai de lin | Kai lan xin | Lai qie li | Li an lin | Li tan | Pu shu | Qi sa | Qing tan wei | Rui an da | Shu an xin | Shu jia qing | Shu li wei | Shu sa lin | Shu ta bi tuo | Sulbactam/ampicil | Tian shu xin | Unasyn | Wei an xin | Xi di lin | Xin an lin | You pu lin | Zhen pai; (CO) Colombia: AMPICILINA + SULBACTAM | Ampicilina sulbactam | Ampicilina y Sulbactam | Ampicilina+sulbactam | Ampidelt | Amplisul | Auropennz | Biosabactam | Fipexiam | Sulamp | Sulbactam + ampicilina | Sulpitam | Sultamicilina | Unasyn | Xilbac; (CR) Costa Rica: AMPICILINA + SULBACTAM | Unasyn; (CZ) Czech Republic: Ampicillin and Sulbactam IBI | Ampicillin/Sulbactam AptaPharma | Bitammon | Unasyn; (DE) Germany: Ampicillin + Sulbactam DeltaSelect | Ampicillin comp | Ampicillin plus sulbactam eberth | Ampicillin Sulb. Teva | Ampicillin und Sulbactam IBI | Ampicillin/Sulbactam Actavis | Ampicillin/sulbactam aurobindo | Ampicillin/Sulbactam Kabi | Ampicillin/sulbactam puren | Unacid; (DO) Dominican Republic: Ampibactam | Fipexiam | Unasyn; (EC) Ecuador: AMPICILINA + SULBACTAM | Ampicilina sodica + sulbactam sodico | Ampicilina sulbactam | Ampicilina y Sulbactam | Ampicilina+sulbactam | Ampicilina/sulbactam biosano | Amplisul | Decilina | Sulbacilin | Sulbam | Sultamix | Unasyn | Xilbac; (EE) Estonia: Alfacid | Ampicillin & sulbactam | Ampisulcillin | Unasyn; (EG) Egypt: Ampictam | Sabect | Sulbin | Synerpen | Ultracillin | Unasyn | Unictam; (ES) Spain: Unasyn; (ET) Ethiopia: Ampicillin and sulbactam |

Ampicillin/sulbactam | Sulbactam/ampicillin; (FR) France: Unacim; (GB) United Kingdom: Dicapen | Unasyn; (GR) Greece: Begalin p | Demotine; (GT) Guatemala: AMPICILINA + SULBACTAM | Ampicilina y Sulbactam | Terabiol; (HK) Hong Kong: Kintamvy | Unasyn; (HU) Hungary: Ampicillin/Sulbactam AptaPharma | Unasyn; (ID) Indonesia: Bactesyn | Cinam | Picyn | Unasyn | Viccillin SX; (IN) India: Ampifect s | Ampirok | Ampitum | Cinclox s | Saltum | Sulbacin | Sulbact; (IT) Italy: Ampicill+Sulb ibi | Bethacil; (JO) Jordan: Unasyn; (JP) Japan: Picillibacta | Pisulcin | Sulbac Ampicil | Sulbacillin | Sulbacsin | Unasyn s | Yucion-s | Yunasupin; (KE) Kenya: Mahacillin s | Sulbacin; (KR) Korea, Republic of: Aerbactam | Ambactam | Ampibactam | Ampitam | Baccillin | Bactacin | Bactamcillin | Lactacillin | Projin | Rukasyun | Ssulbacin | Sulbacillin | Sulbacin | Sulin | Sultam | Synertam | Ubacillin | Ubacsin | Ubactam | Ucilin | Ucilin | Unasyn | Unisulam | Upocin | Uposin | Utacin; (LT) Lithuania: Ampicillin & sulbactam | Ampicillin/sulbactam | Ampiplus | Ampisulcillin | Bestpen s | Bitammon | Unasyn; (LV) Latvia: Ampisid | Sulbacin | Unasyn; (MA) Morocco: Unasyne; (MX) Mexico: Unasyna; (MY) Malaysia: Ampicillin+sulbactam | Auropennz | Sulbacin | Sulbamp | Unasyn; (NG) Nigeria: Unasyn; (NO) Norway: Ampicillin plus sulbactam eberth; (PA) Panama: AMPICILINA + SULBACTAM | Sultamicilina richet | Unasyn; (PE) Peru: AMPICILINA + SULBACTAM | Ampisulkem | Fipexiam | Unasyn; (PH) Philippines: Ambacitam | Ampibax | Ampicare | Ampico Sbt | Ampimax | Ampinex plus | Ampisul | Ampisultam | Amsucillin | Amsulvex | Amsutas | Bactacin | Biocillin plus | Ciltamin | Disulbac | Duobaclin | Gravitam | Linsul | Miasyn | Qualisultam | Sentram | Silgram | Sulbacin | Sulbactizin | Sultacillin | Sybactam | Unasan | Unasyn | Unathix | Unsin | Zulbactam; (PK) Pakistan: Ambac | Ampitam | Sulbacin | Unasyn; (PL) Poland: Ampicillin sulbactam delfarma | Ultraproct | Unasyn; (PR) Puerto Rico: Ampicillin and sulbactam | Unasyn; (PY) Paraguay: Aminoxidin sulbactam | Ampicilina sulbactam eticos | Ampicilina sulbactam gemepe | Ampicilina sulbactam interlabo | Ampicilina sulbactam nortelab | Ampicilina sulbactam vivele | Cilimpil bactam | Klafen | Libractam | Norake ibl; (QA) Qatar: Unasyn; (RO) Romania: Unasyn; (RU) Russian Federation: Amibactam | Ampicillin sulbactam | Ampicillin+sulbactam | Ampisid | Complisan | Duaktilam | Libakcil | Rampinam | Sulacillin | Sulbacin | Sultasin | Unasyn; (SA) Saudi Arabia: Ampiplus | Unasyn | Unictam; (SI) Slovenia: Ampicillin/sulbactam aptapharma | Penactam | Unasyn; (SK) Slovakia: Ampisulcillin | Bitammon | Unasyn; (SV) El Salvador: AMPICILINA + SULBACTAM; (TH) Thailand: Ambacitam | Ampitam | Amsubac | Sulam | Sulbaccin | Unasyn; (TR) Turkey: Devasid | Duobak | Duobaktam | Nobecid | Sulbaksit | Sultasid | Sultibac; (TW) Taiwan: Ambacillin | Amsulber cyh | Ansullina | Subacillin | Sulampi | Unasyn; (UA) Ukraine: Ampicillin+sulbactam | Ampiplus | Ampisid | Ampisulbin | Ampisulcillin | Unasyn; (UG) Uganda: Unictam; (UY) Uruguay: Aminoxidin sulbactam | Ampicilina sulbactam | Ampicilina sulbactam blaufarma uruguay | Ampicilina+sulbactam winpharm | Ampisul | Libractam | Maxicilina Sulbactam | Sulbampicin 1500 a+s | Sulbampicin 3000 a+s | Unasyn; (VE) Venezuela, Bolivarian Republic of: Ampibactan | AMPICILINA + SULBACTAM | Ampicilina sulbactam | Ampicilina+sulbactam | Ampidelt | Ampilab | Ampitren | Ampiwell s | Amtacilyn | Fipexiam | Sentram | Sinif | Sulamp | Sulpexium | Sultamilan | Unacilyn | Unasyn; (VN) Viet Nam: Ama power | Auropennz | Nerusyn | Senitram | Visulin; (ZM) Zambia: Sulbacin

Use of UpToDate is subject to the [Terms of Use](#).

REFERENCES

1. Ailes EC, Gilboa SM, Gill SK, et al; The National Birth Defects Prevention Study. Association between antibiotic use among pregnant women with urinary tract infections in the first trimester and birth defects, National Birth Defects Prevention Study 1997 to 2011. *Birth Defects Res A Clin Mol Teratol*. 2016;106(11):940-949. doi:10.1002/bdra.23570 [PubMed [27891788](#)]
2. Allewelt M, Schüler P, Bölskei PL, Mauch H, Lode H; Study Group on Aspiration Pneumonia. Ampicillin + sulbactam vs clindamycin +/- cephalosporin for the treatment of aspiration pneumonia and primary lung abscess. *Clin Microbiol Infect*. 2004;10(2):163-170. doi:10.1111/j.1469-0691.2004.00774.x [PubMed [14759242](#)]
3. American Academy of Pediatrics (AAP). In: Kimberlin DW, Banerjee R, Barnett ED, Lynfield R, Sawyer MH, eds. *Red Book: 2024-2027 Report of the Committee on Infectious Diseases*. 33rd ed. American Academy of Pediatrics; 2024.
4. American College of Obstetricians and Gynecologists (ACOG) Committee on Obstetric Practice. Committee opinion No. 712: Intrapartum management of intraamniotic infection. *Obstet Gynecol*. 2017;130(2):e95-e101. doi:10.1097/AOG.0000000000002236 [PubMed [28742677](#)]
5. American College of Obstetricians and Gynecologists (ACOG) Committee on Practice Bulletins-Obstetrics. ACOG Practice Bulletin No. 199: Use of prophylactic antibiotics in labor and delivery. *Obstet Gynecol*. 2018;132(3):e103-e119. [PubMed [30134425](#)]
6. Aronoff SC, Scoles PV, Makley JT, Jacobs MR, Blumer JL, Kalamchi A. Efficacy and safety of sequential treatment with parenteral sulbactam/ampicillin and oral sulfamycin for skeletal infections in children. *Rev Infect Dis*. 1986;8(suppl 5):S639-S643. doi:10.1093/clinids/8.supplement_5.s639 [PubMed [3026018](#)]
7. Assimakopoulos SF, Karamouzou V, Lefkaditi A, et al. Triple combination therapy with high-dose ampicillin/sulbactam, high-dose tigecycline and colistin in the treatment of ventilator-associated pneumonia caused by pan-drug resistant *Acinetobacter baumannii*: a case series study. *Infez Med*. 2019;27(1):11-16. [PubMed [30882373](#)]
8. Baddour LM, Harper M. Animal bites (dogs, cats, and other mammals): Evaluation and management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 24, 2024a.
9. Baddour LM, Harper M. Human bites: evaluation and management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 24, 2024b.
10. Baddour LM, Wilson WR, Bayer AS, et al; American Heart Association Committee on Rheumatic Fever, Endocarditis, and Kawasaki Disease of the Council on Cardiovascular Disease in the Young, Council on Clinical Cardiology, Council on Cardiovascular Surgery and Anesthesia, and Stroke Council. Infective endocarditis in adults: diagnosis, antimicrobial therapy, and management of complications: a scientific statement for healthcare professionals from the American Heart Association. *Circulation*. 2015;132(15):1435-1486. [PubMed [26373316](#)]
11. Baltimore RS, Gewitz M, Baddour LM, et al; American Heart Association Rheumatic Fever, Endocarditis, and Kawasaki Disease Committee of the Council on Cardiovascular Disease in the Young and the Council on Cardiovascular and Stroke Nursing. Infective endocarditis in childhood: 2015 update: a scientific statement from the American Heart Association. *Circulation*. 2015;132(15):1487-1515. doi:10.1161/CIR.0000000000000298 [PubMed [26373317](#)]
12. Beigi RH. Tubo-ovarian abscess: Management and complications. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed September 8, 2020.

13. Beganovic M, Daffinee KE, Luther MK, LaPlante KL. Minocycline alone and in combination with polymyxin b, meropenem, and sulbactam against carbapenem-susceptible and -resistant *Acinetobacter baumannii* in an in vitro pharmacodynamic model. *Antimicrob Agents Chemother*. 2021;65(3):e01680-20. doi:10.1128/AAC.01680-20 [PubMed [33318006](#)]
14. Berrios-Torres SI, Umscheid CA, Bratzler DW, et al; Healthcare Infection Control Practices Advisory Committee. Centers for Disease Control and Prevention guideline for the prevention of surgical site infection, 2017 [published correction appears in *JAMA Surg*. 2017;152(8):803]. *JAMA Surg*. 2017;152(8):784-791. doi:10.1001/jamasurg.2017.0904 [PubMed [28467526](#)]
15. Betrosian AP, Frantzeskaki F, Xanthaki A, Douzinas EE. Efficacy and safety of high-dose ampicillin/sulbactam vs. colistin as monotherapy for the treatment of multidrug resistant *Acinetobacter baumannii* ventilator-associated pneumonia. *J Infect*. 2008;56(6):432-436. doi:10.1016/j.jinf.2008.04.002 [PubMed [18501431](#)]
16. Bilbao-Meseguer I, Rodríguez-Gascón A, Barrasa H, Isla A, Solinís MÁ. Augmented renal clearance in critically ill patients: a systematic review. *Clin Pharmacokinet*. 2018;57(9):1107-1121. doi:10.1007/s40262-018-0636-7 [PubMed [29441476](#)]
17. Blackwell BG, Leggett JE, Johnson CA, Zimmerman SW, Craig WA. Ampicillin and sulbactam pharmacokinetics and pharmacodynamics in continuous ambulatory peritoneal dialysis (CAPD). *Perit Dial Int*. 1990;10(3):221-226. [PubMed [2099158](#)]
18. Bookstaver PB, Bland CM, Griffin B, Stover KR, Eiland LS, McLaughlin M. A review of antibiotic use in pregnancy. *Pharmacotherapy*. 2015 ;35(11):1052-1062. doi:10.1002/phar.1649 [PubMed [26598097](#)]
19. Bower WA, Yu Y, Person MK, et al. CDC guidelines for the prevention and treatment of anthrax, 2023. *MMWR Recomm Rep*. 2023;72(6):1-47. doi:10.15585/mmwr.rr7206a1 [PubMed [37963097](#)]
20. Bradley JS, Nelson JD, Barnett ED, et al, eds. *Nelson's Pediatric Antimicrobial Therapy*. 30th ed. American Academy of Pediatrics; 2024.
21. Bratzler DW, Dellinger EP, Olsen KM, et al; American Society of Health-System Pharmacists; Infectious Diseases Society of America; Surgical Infection Society; Society for Healthcare Epidemiology of America. Clinical practice guidelines for antimicrobial prophylaxis in surgery. *Am J Health Syst Pharm*. 2013;70(3):195-283. [PubMed [23327981](#)]
22. Burkart JM. Microbiology and therapy of peritonitis in peritoneal dialysis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 21, 2022.
23. Chamberlain A, White S, Bawdon R, Thomas S, Larsen B. Pharmacokinetics of ampicillin and sulbactam in pregnancy. *Am J Obstet Gynecol*. 1993;168(2):667-673.
24. Chan JD, Graves JA, Dellit TH. Antimicrobial treatment and clinical outcomes of carbapenem-resistant *Acinetobacter baumannii* ventilator-associated pneumonia. *J Intensive Care Med*. 2010;25(6):343-348. doi:10.1177/0885066610377975 [PubMed [20837632](#)]
25. Chen KT. Postpartum endometritis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed March 10, 2021.
26. Chow AW. Complications, diagnosis, and treatment of odontogenic infections. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 9, 2022.
27. Chow AW, Benninger MS, Brook I, et al; Infectious Diseases Society of America. IDSA clinical practice guideline for acute bacterial rhinosinusitis in children and adults. *Clin Infect Dis*. 2012;54(8):e72-e112. [PubMed [22438350](#)]

28. Cisneros JM, Reyes MJ, Pachón J, et al. Bacteremia due to *Acinetobacter baumannii*: epidemiology, clinical findings, and prognostic features. *Clin Infect Dis*. 1996;22(6):1026-1032. doi:10.1093/clinids/22.6.1026 [PubMed [8783704](#)]
29. Crider KS, Cleves MA, Reefhuis J, Berry RJ, Hobbs CA, Hu DJ. Antibacterial medication use during pregnancy and risk of birth defects: National Birth Defects Prevention Study. *Arch Pediatr Adolesc Med*. 2009 ;163(11):978-985. doi:10.1001/archpediatrics.2009.188 [PubMed [19884587](#)]
30. Damkier P, Brønnicke LMS, Korch-Frandsen JFB, Broe A. In utero exposure to antibiotics and risk of congenital malformations: a population-based study. *Am J Obstet Gynecol*. 2019;221(6):648.e1-648.e15. doi:10.1016/j.ajog.2019.06.050 [PubMed [31260651](#)]
31. Delgado V, Ajmone Marsan N, de Waha S, et al. 2023 ESC guidelines for the management of endocarditis. *Eur Heart J*. 2023;44(39):3948-4042. doi:10.1093/eurheartj/ehad193 [PubMed [37622656](#)]
32. Expert opinion based on Jaruratanasirikul 2019.
33. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
34. Foulds G. Pharmacokinetics of sulbactam/ampicillin in humans: a review. *Rev Infect Dis*. 1986;8(suppl 5):S503-S510. [PubMed [3025997](#)]
35. Galante D, Esposito S, Barba D, Ruffilli MP. Clinical efficacy and safety of sulbactam/ampicillin in patients suffering from chronic liver disease. *J Antimicrob Chemother*. 1987;19(4):527-532. doi:10.1093/jac/19.4.527 [PubMed [3034850](#)]
36. Gall S, Koukol DH. Ampicillin/sulbactam vs. clindamycin/gentamicin in the treatment of postpartum endometritis. *J Reprod Med*. 1996;41(8):575-580. [PubMed [8866384](#)]
37. Geckler RW. A comparison of ampicillin/sulbactam and cefuroxime in the treatment of patients with bacterial infections of the lower respiratory tract. *Clin Ther*. 1994;16(4):662-672. [PubMed [7982254](#)]
38. Gilad J, Carmeli Y. Treatment options for multidrug-resistant *Acinetobacter* species. *Drugs*. 2008;68(2):165-189. [PubMed [18197724](#)]
39. Gkentzi D, Tsintoni A, Christopoulou I, et al. Extensively-drug resistant *Acinetobacter baumannii* bacteremia in neonates: effective treatment with the combination of colistin and ampicillin/sulbactam. *J Chemother*. 2020;32(2):103-106. doi:10.1080/1120009X.2020.1716478 [PubMed [31992156](#)]
40. Halstenson CE, Wong MO, Herman CS, et al. Effect of concomitant administration of piperacillin on the dispositions on isepamicin and gentamicin in patients with end-stage renal disease. *Antimicrob Agents Chemother*. 1992;36(9):1832-1836. [PubMed [1416875](#)]
41. Harkless L, Boghossian J, Pollak R, et al. An open-label, randomized study comparing efficacy and safety of intravenous piperacillin/tazobactam and ampicillin/sulbactam for infected diabetic foot ulcers. *Surg Infect (Larchmt)*. 2005;6(1):27-40. doi:10.1089/sur.2005.6.27 [PubMed [15865549](#)]
42. Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. *Pharmacotherapy*. 2009;29(5):562-577. [PubMed [19397464](#)]
43. Housman ST, Hagihara M, Nicolau DP, Kuti JL. In vitro pharmacodynamics of human-simulated exposures of ampicillin/sulbactam, doripenem and tigecycline alone and in combination against multidrug-resistant *Acinetobacter baumannii*. *J Antimicrob Chemother*. 2013;68(10):2296-2304. doi:10.1093/jac/dkt197 [PubMed [23710070](#)]

44. Jaruratanasirikul S, Nitchot W, Wongpoowarak W, Samaeng M, Nawakittrangsarn M. Population pharmacokinetics and Monte Carlo simulations of sulbactam to optimize dosage regimens in patients with ventilator-associated pneumonia caused by *Acinetobacter baumannii*. *Eur J Pharm Sci*. 2019;136:104940. doi:10.1016/j.ejps.2019.05.018 [PubMed [31132402](#)]
45. Jellison TK, McKinnon PS, Rybak MJ. Epidemiology, resistance, and outcomes of *Acinetobacter baumannii* bacteremia treated with imipenem-cilastatin or ampicillin-sulbactam. *Pharmacotherapy*. 2001;21(2):142-148. [PubMed [11213849](#)]
46. Jusko WJ, Lewis GP, Schmitt GW. Ampicillin and hetacillin pharmacokinetics in normal and anephric subjects. *Clin Pharmacol Ther*. 1973;14(1):90-99. doi:10.1002/cpt197314190 [PubMed [4345622](#)]
47. Kalil AC, Metersky ML, Klompas M, et al. Executive summary: management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clin Infect Dis*. 2016;63(5):575-582. [PubMed [27521441](#)]
48. Kanafani ZA, Kanj SS. *Acinetobacter* infection: treatment and prevention. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 14, 2019.
49. Kanra G, Seçmeer G, Ozen H, Ceyhan M, Ecevit Z. Comparative therapeutic results of penicillin plus chloramphenicol versus ampicillin plus sulbactam in childhood meningococemia. *Turk J Pediatr*. 1993;35(2):87-91. [PubMed [8249199](#)]
50. Klompas M. Aspiration pneumonia in adults. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed July 1, 2024.
51. Kulhanjian J, Dunphy MG, Hamstra S, et al. Randomized comparative study of ampicillin/sulbactam vs ceftriaxone for treatment of soft tissue and skeletal infections in children. *Pediatr Infect Dis J*. 1989;8(9):605-610. [PubMed [2677956](#)]
52. Lam MF, Tang BS, Tse KC, Chan TM, Lai KN. Ampicillin-sulbactam and amikacin used as second-line antibiotics for patients with culture-negative peritonitis. *Perit Dial Int*. 2008;28(5):540-542. [PubMed [18708550](#)]
53. Lamont HF, Blogg HJ, Lamont RF. Safety of antimicrobial treatment during pregnancy: a current review of resistance, immunomodulation and teratogenicity. *Expert Opin Drug Saf*. 2014;13(12):1569-1581. doi:10.1517/14740338.2014.939580 [PubMed [25189188](#)]
54. Levin AS, Levy CE, Manrique AE, Medeiros EA, Costa SF. Severe nosocomial infections with imipenem-resistant *Acinetobacter baumannii* treated with ampicillin/sulbactam. *Int J Antimicrob Agents*. 2003;21(1):58-62. [PubMed [12507838](#)]
55. Li PK, Chow KM, Cho Y, et al. ISPD peritonitis guideline recommendations: 2022 update on prevention and treatment. *Perit Dial Int*. 2022;42(2):110-153. doi:10.1177/08968608221080586 [PubMed [35264029](#)]
56. Lorenzen JM, Broll M, Kaefer V, et al. Pharmacokinetics of ampicillin/sulbactam in critically ill patients with acute kidney injury undergoing extended dialysis. *Clin J Am Soc Nephrol*. 2012;7(3):385-390. doi:10.2215/CJN.05690611 [PubMed [2223613](#)]
57. Maberry MC, Trimmer KJ, Bawdon RE, Sobhi S, Dax JB, Gilstrap LC 3rd. Antibiotic concentration in maternal blood, cord blood and placental tissue in women with chorioamnionitis. *Gynecol Obstet Invest*. 1992;33(3):185-186. doi:10.1159/000294878 [PubMed [1612532](#)]
58. Majcher-Peszynska J, Loebermann M, Klammt S, et al. CAPNETZ Study Group. Ampicillin/sulbactam in elderly patients with community-acquired pneumonia. *Infection*. 2014;42(1):79-87. [PubMed [23904004](#)]

59. Makris D, Petinaki E, Tsolaki V, et al. Colistin versus colistin combined with ampicillin-sulbactam for multiresistant *Acinetobacter baumannii* ventilator-associated pneumonia treatment: an open-label prospective study. *Indian J Crit Care Med*. 2018;22(2):67-77. doi:10.4103/ijccm.IJCCM_302_17 [PubMed [29531445](#)]
60. Mancino AT, Lalani T. Wound infection following repair of abdominal wall hernia. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 21, 2022.
61. Manzoni P, Esposito S, Gallo E, et al. Switch therapy in full-term neonates with presumed or proven bacterial infection. *J Chemother*. 2009;21(1):68-73. [PubMed [19297276](#)]
62. Marumo S, Teranishi T, Higami Y, Koshimo Y, Kiyokawa H, Kato M. Effectiveness of azithromycin in aspiration pneumonia: a prospective observational study. *BMC Infect Dis*. 2014;14:685. doi:10.1186/s12879-014-0685-y [PubMed [25491126](#)]
63. Mazuski JE, Tessier JM, May AK, et al. The Surgical Infection Society (SIS) revised guidelines on the management of intra-abdominal infection. *Surg Infect (Larchmt)*. 2017;18(1):1-76. doi:10.1089/sur.2016.261 [PubMed [28085573](#)]
64. Meaney-Delman D, Zotti ME, Creanga AA, et al. Special considerations for prophylaxis for and treatment of anthrax in pregnant and postpartum women. *Emerg Infect Dis*. 2014;20(2):e130611. doi:10.3201/eid2002.130611 [PubMed [24457117](#)]
65. Mermel LA, Allon M, Bouza E, et al. Clinical practice guidelines for the diagnosis and management of intravascular catheter-related infection: 2009 update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2009;49(1):1-45. [PubMed [19489710](#)]
66. Metlay JP, Waterer GW, Long AC, et al. Diagnosis and treatment of adults with community-acquired pneumonia. An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America. *Am J Resp Crit Care Med*. 2019;200(7):e45-e67. doi:10.1164/rccm.201908-1581ST [PubMed [31573350](#)]
67. Meyers BR, Wilkinson P, Mendelson MH, et al. Pharmacokinetics of ampicillin-sulbactam in healthy elderly and young volunteers. *Antimicrob Agents Chemother*. 1991;35(10):2098-2101. [PubMed [1759832](#)]
68. Miller WR. Treatment of enterococcal infections. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 14, 2019.
69. Moehring R, Anderson DJ. Gram-negative bacillary bacteremia in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 12, 2019.
70. Mosaed R, Haghighi M, Kouchak M, et al. Interim study: comparison of safety and efficacy of levofloxacin plus colistin regimen with levofloxacin plus high dose ampicillin/sulbactam infusion in treatment of ventilator-associated pneumonia due to multidrug resistant *Acinetobacter*. *Iran J Pharm Res*. 2018;17(suppl 2):206-213. [PubMed [31011353](#)]
71. Muanda FT, Sheehy O, Bérard A. Use of antibiotics during pregnancy and the risk of major congenital malformations: a population based cohort study. *Br J Clin Pharmacol*. 2017a;83(11):2557-2571. doi:10.1111/bcp.13364 [PubMed [28722171](#)]
72. Muanda FT, Sheehy O, Bérard A. Use of antibiotics during pregnancy and risk of spontaneous abortion. *CMAJ*. 2017b;189(17):E625-E633. doi:10.1503/cmaj.161020 [PubMed [28461374](#)]
73. Nahata MC, Vashi VI, Swanson RN, Messig MA, Chung M. Pharmacokinetics of ampicillin and sulbactam in pediatric patients. *Antimicrob Agents Chemother*. 1999;43(5):1225-1229. doi:10.1128/AAC.43.5.1225 [PubMed [10223940](#)]
74. Refer to manufacturer's labeling.

75. Rho SP, Jones A, Woo M, et al. Single dose pharmacokinetics of intravenous ampicillin plus sulbactam in healthy elderly and young subjects. *J Antimicrob Chemother.* 1989;24(4):573-580. [PubMed [2613605](#)]
76. Rodríguez WJ, Khan WN, Puig J, et al. Sulbactam/ampicillin vs. chloramphenicol/ampicillin for the treatment of meningitis in infants and children. *Rev Infect Dis.* 1986;8(suppl 5):S620-S629. doi:10.1093/clinids/8.supplement_5.s620 [PubMed [3026015](#)]
77. Rossoff LJ, Hilton E, Smith C, Isenberg HD, Simme MM. Intravenous ampicillin/sulbactam versus cefuroxime axetil in the treatment of patients hospitalized with community-acquired lower respiratory tract infections. *Curr Ther Res.* 1995;56(9):852-862.
78. Saavedra-Lozano J, Falup-Pecurariu O, Faust SN, et al. Bone and joint infections. *Pediatr Infect Dis J.* 2017;36(8):788-799. doi:10.1097/INF.0000000000001635 [PubMed [28708801](#)]
79. Senneville É, Albalawi Z, van Asten SA, et al. IWGDF/IDSA guidelines on the diagnosis and treatment of diabetes-related foot infections (IWGDF/IDSA 2023). *Diabetes Metab Res Rev.* 2024;40(3):e3687. doi:10.1002/dmrr.3687 [PubMed [37779323](#)]
80. Smolyakov R, Borer A, Riesenberger K, et al. Nosocomial multi-drug resistant *Acinetobacter baumannii* bloodstream infection: risk factors and outcome with ampicillin-sulbactam treatment. *J Hosp Infect.* 2003;54(1):32-38. [PubMed [12767844](#)]
81. Solomkin JS, Mazuski JE, Bradley JS, et al. Diagnosis and Management of Complicated Intra-Abdominal Infections in Adults and Children: Guidelines by the Surgical Infection Society and the Infectious Diseases Society of America. *Clin Infect Dis.* 2010;50(2):133-164. [PubMed [20034345](#)]
82. Stevens DL, Bisno AL, Chambers HF, et al. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America. *Clin Infect Dis.* 2014;59(2):e10-e52. doi: 10.1093/cid/ciu444. [PubMed [24973422](#)]
83. Sutton AM, Turner TL, Cockburn F, et al. Pharmacokinetic study of sulbactam and ampicillin administered concomitantly by intraarterial or intravenous infusion in the newborn. *Rev Infect Dis.* 1986;8(suppl 5):518-522. [PubMed [3025999](#)]
84. Syriopoulou V, Bitsi M, Theodoridis C, et al. Clinical Efficacy of Sulbactam/ampicillin in pediatric infections caused by ampicillin-resistant or penicillin-resistant organisms. *Rev Infect Dis.* 1986;8(suppl 5):630-633.
85. Tamma PD, Heil EL, Justo JA, Mathers AJ, Satlin MJ, Bonomo RA. Infectious Diseases Society of America antimicrobial-resistant treatment guidance: gram-negative bacterial infections. *Clin Infect Dis.* Published online July 12, 2024.
86. Tamma PD, Heil EL, Justo JA, Mathers AJ, Satlin MJ, Bonomo RA. Infectious Diseases Society of America 2024 guidance on the treatment of antimicrobial-resistant gram-negative infections. *Clin Infect Dis.* Published online August 7, 2024. <https://www.idsociety.org/practice-guideline/amr-guidance/>. doi:10.1093/cid/ciae403 [PubMed [39108079](#)]
87. Tita ATN. Clinical chorioamnionitis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 24, 2024.
88. Trotman RL, Williamson JC, Shoemaker DM, Salzer WL. Antibiotic dosing in critically ill adult patients receiving continuous renal replacement therapy. *Clin Infect Dis.* 2005;41(8):1159-1166. doi:10.1086/444500 [PubMed [16163635](#)]
89. Tunkel AR, Hartman BJ, Kaplan SL, et al. Practice guidelines for the management of bacterial meningitis. *Clin Infect Dis.* 2004;39(9):1267-1284. doi:10.1086/425368 [PubMed [15494903](#)]

90. Tunkel AR, Hasbun R, Bhimraj A, et al. 2017 Infectious Diseases Society of America's clinical practice guidelines for healthcare-associated ventriculitis and meningitis. *Clin Infect Dis*. 2017;64(6):e34-e65. doi:10.1093/cid/ciw861 [PubMed [28203777](#)]
91. Udy AA, Roberts JA, Boots RJ, Paterson DL, Lipman J. Augmented renal clearance: implications for antibacterial dosing in the critically ill. *Clin Pharmacokinet*. 2010;49(1):1-16. doi:10.2165/11318140-000000000-00000 [PubMed [20000886](#)]
92. Unasyn (ampicillin sodium and sulbactam sodium) [prescribing information]. New York, NY: Roerig; November 2018.
93. Unasyn (ampicillin sodium and sulbactam sodium) [prescribing information]. New York, NY: Roerig; October 2024.
94. Unasyn (ampicillin sodium and sulbactam sodium) [prescribing information]. New York, NY: Pfizer Inc; November 2025.
95. Unasyn pharmacy bulk package (ampicillin sodium and sulbactam sodium) [prescribing information]. New York, NY: Pfizer Inc; November 2025.
96. Wilson KH, Cardile AP. Treatment of anthrax. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 29, 2024.
97. Workowski KA, Bachmann LH, Chan PA, et al. Sexually transmitted infections treatment guidelines, 2021. *MMWR Recomm Rep*. 2021;70(4):1-187. doi:10.15585/mmwr.rr7004a1 [PubMed [34292926](#)]
98. Workowski KA, Bolan GA; Centers for Disease Control and Prevention. Sexually transmitted diseases treatment guidelines, 2015. *MMWR Recomm Rep*. 2015;64(RR-03):1-137. [PubMed [26042815](#)]
99. World Health Organization (WHO). Breastfeeding and maternal medication, recommendations for drugs in the eleventh WHO model list of essential drugs. Published 2002. Available at <https://apps.who.int/iris/handle/10665/62435>.
100. Wright N, Wise R. The elimination of sulbactam alone and combined with ampicillin in patients with renal dysfunction. *J Antimicrob Chemother*. 1983;11(6):583-587. doi:10.1093/jac/11.6.583 [PubMed [6309731](#)]
101. Wynn RL, Bergman SA. Antibiotics and their use in the treatment of orofacial infections, part I and part II. *Gen Dent*. 1994;42(5):398-402, 498-502. [PubMed [7489869](#)]
102. Yahav D, Franceschini E, Koppel F, et al. Bacteremia Duration Study Group. Seven versus fourteen days of antibiotic therapy for uncomplicated gram-negative bacteremia: a noninferiority randomized controlled trial. *Clin Infect Dis*. 2019;69(7):1091-1098. doi:10.1093/cid/ciy1054 [PubMed [30535100](#)]
103. Ye JJ, Lin HS, Yeh CF, et al. Tigecycline-based versus sulbactam-based treatment for pneumonia involving multidrug-resistant *Acinetobacter calcoaceticus*-*Acinetobacter baumannii* complex. *BMC Infect Dis*. 2016;16:374. doi:10.1186/s12879-016-1717-6 [PubMed [27496018](#)]
104. Yokoyama Y, Matsumoto K, Ikawa K, et al. Pharmacokinetic/pharmacodynamic evaluation of sulbactam against *Acinetobacter baumannii* in in vitro and murine thigh and lung infection models. *Int J Antimicrob Agents*. 2014;43(6):547-552. doi:10.1016/j.ijantimicag.2014.02.012 [PubMed [24796218](#)]
105. Yokoyama Y, Matsumoto K, Ikawa K, Watanabe E, Morikawa N, Takeda Y. Population pharmacokinetic-pharmacodynamic target attainment analysis of sulbactam in patients with impaired renal function: dosing considerations for *Acinetobacter baumannii* infections. *J Infect Chemother*. 2015;21(4):284-289. doi:10.1016/j.jiac.2014.12.005 [PubMed [25638291](#)]

106. Zalts R, Neuberger A, Hussein K, et al. Treatment of carbapenem-resistant *Acinetobacter baumannii* ventilator-associated pneumonia: retrospective comparison between intravenous colistin and intravenous ampicillin-sulbactam. *Am J Ther.* 2016;23(1):e78-e85. doi:10.1097/MJT.0b013e3182a32df3 [PubMed [24263165](#)]

Topic 8745 Version 494.0

